

AUTOMATED WILDFIRE HAZARD MODELING USING GOOGLE EARTH ENGINE: A
CASE STUDY OF THE 2024 LINE FIRE

By

HEATHER MARIE ROBBINS

MASTER OF SCIENCE
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To my Mom and Dad.

ACKNOWLEDGMENTS

I thank my parents for all their continuous support. I am eternally grateful for your love and inspiration to continue my educational journey.

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LIST OF ABBREVIATIONS

AOI	Area of Interest
dNBR	Differenced Normalized Burn Ratio
FWI	Fire Weather Index
GAMA	GIS Automated Multicriteria Assessment
GEE	Google Earth Engine
GIS	Geographic Information Science
MCDA	Multi-Criteria Decision Analysis
MTBS	Monitoring Trends in Burn Severity
NBR	Normalized Burn Ratio
NDVI	Normalized Difference Vegetation Index
QGIS	Quantum GIS
USGS	United States Geological Survey
UTM	Universal Transverse Mercator
VPD	Vapor Pressure Deficit
WFPI	Wildland Fire Potential Index
WHI	Wildfire Hazard Index

ABSTRACT

Timely wildfire hazard assessments are critical for disaster preparedness and response. Current approaches to susceptibility modeling are tedious, labor and data-intensive, and often reliant on proprietary software. These limitations can lead to inconsistent workflows, reduced reproducibility, and delays in delivering actionable information for mitigation strategies. This project delivers an automated, open-source workflow designed to estimate wildfire hazard for any study area within the United States. Using Google Earth Engine, the model integrates meteorological, vegetation, and terrain variables obtained from publicly accessible geospatial data. Users can modify the area of interest, define start and end dates for the analysis period, and adjust coefficient weights for model variables to better reflect environmental and temporal dynamics surrounding wildfire seasons. The workflow is tested using the 2024 Line Fire in the San Bernardino National Forest. The primary result of the project is a comprehensive script that generates a Wildfire Hazard Index raster. Built-in helper functions provide flexibility and adaptation for different regions and timeframes, supporting resource managers and emergency planners. By automating laborious tasks of data acquisition and pre-processing, the workflow allows producing consistent and repeatable hazard assessments that can support better decision-making.

Keywords: Wildfire, Open Source, GIS Analysis, Google Earth Engine, Automation

ETHICS STATEMENT

Geographic Information Systems and Technology (GIST) have changed how people perceive and interact with the world. Today, geospatial technologies directly shape decisions in the environmental, social, and economic sectors. Innovation springs from the convergence of technologies. This process can empower or exclude users. The outcome depends on how GIS tools are designed and delivered.

As developers and cartographers, we must preserve ethical standards. These include equity, inclusion, transparency, and proper attribution. The GIS Certification Institute Code of Ethics calls for responsible and professional conduct in all geospatial work. Effective GIS practice requires both technical skill and awareness of how data are collected, processed, and communicated. Practitioners should also consider the social impacts of their work.

The main objective of this project is to develop an effective wildfire-susceptibility assessment model using accessible, open-source tools. The project also demonstrates the capabilities of cloud computing with open-access resources, lowering barriers to entry and promoting collaboration in wildfire science and risk assessment. Additionally, a core ethical aim is to improve modeling workflows and best practices.

This study acknowledges the limitations of its data and methods. All datasets come from publicly available sources and are accepted based on the credibility of their publishers. These limitations are clearly documented. Metadata tables reflect the data available at the time of analysis. All modeling methods are fully cited, supporting openness and reproducibility.

This model is not intended to serve as the sole basis for decision-making. Instead, it functions as a proof of concept for automated, cloud-based geospatial analysis. The results may support land management, emergency planning, and public safety strategies. However, they are best used to support further research and methodological refinement. This work also shows the value of open-source tools in expanding access to geospatial analysis.

Overall, this research was conducted with a responsible approach to geospatial analysis. It recognizes both the capabilities and limitations of GIS technologies. This project supports equitable and resilient geospatial solutions by offering accessibility, transparency, and accountability. Looking forward, this work encourages continuous advancement, collaboration, and ethical practice. Ultimately, by demonstrating how open-source tools promote positive change within the field, this project sets a clear pathway for future efforts to build a more just and sustainable geospatial landscape.

INTRODUCTION

Wildfire ignition results from interactions between fuel, heat, and oxygen. Spread, severity, and containment depend on environmental conditions. These events impact ecosystems, human health, and economies. Greenhouse gases from fires contribute to atmospheric warming, creating a feedback loop where climate change heightens fire risk.

Globally, overall fire frequency and total area burned have decreased. However, extreme fire events have increased 2.2-fold over the past two decades (Cunningham et al., 2024). A University of Colorado study focuses on “fast fires,” driven by high winds, dry vegetation, and elevated temperatures. These rapid-fire events account for approximately 89% of wildfire-related damages and 66% of associated fatalities, totaling more than \$18 billion in costs (NASA Earth Observatory, 2024). The southwestern U.S. is especially vulnerable to drought and high winds.

As climate change worsens, landscapes become more prone to rapid fires. Over 80% of U.S. wildfires are human-caused. Projections show fire seasons will lengthen and intensify, possibly lasting year-round (Center for Climate and Energy Solutions, 2026). Due to unpredictability and extended seasons, timely hazard assessments are essential for emergency response and mitigation.

Modeling with Remote Sensing

To better understand wildfire hazards, remote sensing technologies provide key insights into wildfire research by capturing various climate and geographic conditions over time. In particular, multispectral imagery enables us to examine dynamic wildfire attributes, including the spatial distribution of fire, vegetation health, moisture content,

and the severity of burned landscapes. Such observations are commonly gathered from spectral indices—using the data collected by highly advanced sensors.

The Normalized Difference Vegetation Index (NDVI) measures vegetation health and productivity using near-infrared and red-light reflectance. Stressed and/or dry vegetation provides more combustible fuel; therefore, NDVI is frequently used as an indicator of wildfire susceptibility (Chuvieco et al., 2004). The Normalized Burn Ratio (NBR) is another standardized index that highlights burned areas by calculating the contrast between near-infrared and shortwave-infrared reflectance. The difference between pre-fire and post-fire NBR values, the differenced Normalized Burn Ratio (dNBR), is widely used to quantify burn severity and examine the ecological impacts of wildfire events (Chen et al., 2024).

The LANDFIRE and Monitoring Trends in Burn Severity (MTBS) Programs are examples of collaborative projects providing spatial wildfire datasets. Using Landsat satellite data, MTBS produces burn severity maps and fire perimeter products that inform wildfire management policies. In wildfire studies, most remote sensing applications focus on fire detection or burn severity, and fewer target wildfire hazard estimation.

Historical Wildfire Workflows

Many studies have analyzed wildfire behavior and risk globally, resulting in various fire danger indices. However, no single standard algorithm exists for assessing wildfire hazards. The lack of a standardized process is likely due to varying climates and geographic conditions worldwide. Most wildfire susceptibility models are developed using a multi-criteria decision analysis (MCDA) framework. In this type of framework,

input variables are normalized to a common scale and combined via a weighted linear combination to produce a continuous hazard surface.

Model validation involves comparing hazard estimates to known fire locations. The Mann–Whitney U test, a nonparametric method, assesses the significance of differences between groups when data are not normally distributed (Chuvieco et al., 2009).

The primary variables selected as inputs across established frameworks are seasonal climate and vegetation conditions. The Pan-European wildfire risk assessment model, supported by the European Union, measures wildfire risk as the interaction of hazard, exposure, and vulnerability (Jacome et al., 2022). Within the hazard component of this assessment model, variables such as fuel moisture content, terrain, wind, humidity, and vegetation type are used to model expected fire behavior. The United States Geological Survey (USGS) developed the Wildland Fire Potential Index (WFPI), which integrates wind, vegetation condition, atmospheric moisture, precipitation, and temperature.

This project focuses on the *hazard* component of wildfire risk, examining how regional climate and geographic factors relate using the San Bernardino National Forest as a case study.

Established frameworks of wildfire susceptibility modeling often require analysts to collect and process large volumes of geospatial data from multiple sources and formats. This work demands comprehensive knowledge of GIS operations. Additionally, producing impactful outputs that support mitigation and response planning often

requires access to commercial software. Therefore, the nature of current methods suggests the need for updated, simplified hazard modeling approaches.

Automation, Open-Source Tools and Google Earth Engine

Existing wildfire hazard modeling tools in web-based GIS environments are limited. Publicly available climate risk assessments rely heavily on commercial platforms such as ArcGIS, developed by Esri. They typically require specialized knowledge of GIS workflows and coding environments. In the relatively few cases where open-source models exist, they are often constrained to specific geographic regions and lack flexibility for broader application.

The CLIMAAX Project (CLIMAt risk and vulnerability Assessment framework and toolboX) is one example of an open-source initiative providing climate risk analysis. The project provides Python scripts and step-by-step tutorials that enable users to assess snow, wind, drought, wildfire, and coastal flooding (Climaax, 2026). The group behind the project also encourages collaboration through an extensive GitHub repository. However, limitations consist of local data downloads, familiarity with Python programming, and geographic constraints to European regions.

Automated workflows boost reproducibility and efficiency by streamlining complex tasks. For example, the GIS Automated Multicriteria Assessment (GAMA) method lets users set parameters for susceptibility models, automating the standardization, weighting, and aggregation of criteria (Domazetović et al., 2019). However, this approach has typically required commercial software licenses and hardware capable of processing large geospatial datasets, prompting the exploration of more accessible platforms.

Cloud computing platforms offer an alternative approach, enabling large-scale geospatial analysis without requiring data storage or specialized software. **Google Earth Engine** (GEE) is an open-source geospatial platform that provides access to petabytes of satellite imagery and environmental data (Cardille et al., 2023). Through its interactive development environment, users can access, analyze, and export composite images. In this context, GEE empowers end users to perform large-scale data analysis directly within Google's cloud infrastructure, from any machine with internet access.

Project Goals

Building on these capabilities, the primary objective of this project is to develop an automated wildfire hazard modeling workflow. More specifically, the study develops a Wildfire Hazard Index (WHI) raster output from environmental variables representing pre-season conditions within a defined time period. Given the capabilities of cloud computing and the need for accessible wildfire hazard modeling, the following research question is addressed:

How effectively can an automated, open-source workflow estimate wildfire hazard when evaluated against observed fire perimeters?

Implementation of a repeatable workflow in Google Earth Engine eliminates the need for large data downloads and streamlines geospatial analysis using cloud-based data. Ultimately, this project shows that automated, open-source workflows can provide efficient hazard models while improving accessibility and reproducibility for future wildfire research and management applications.

METHODS

Study Area

The San Bernardino National Forest covers over 820,000 acres (3,333 km²) in Riverside and San Bernardino Counties, Southern California. The forest is divided into a northern and a southern portion, separated by a corridor along I-10 near Banning and Cabazon. Known as the “most urban” forest, it contains mountain ranges, shrublands, and urban communities. The forest lies between the Mojave Desert and the Inland Empire. The landscape is prone to drought, high winds, and extreme heat.

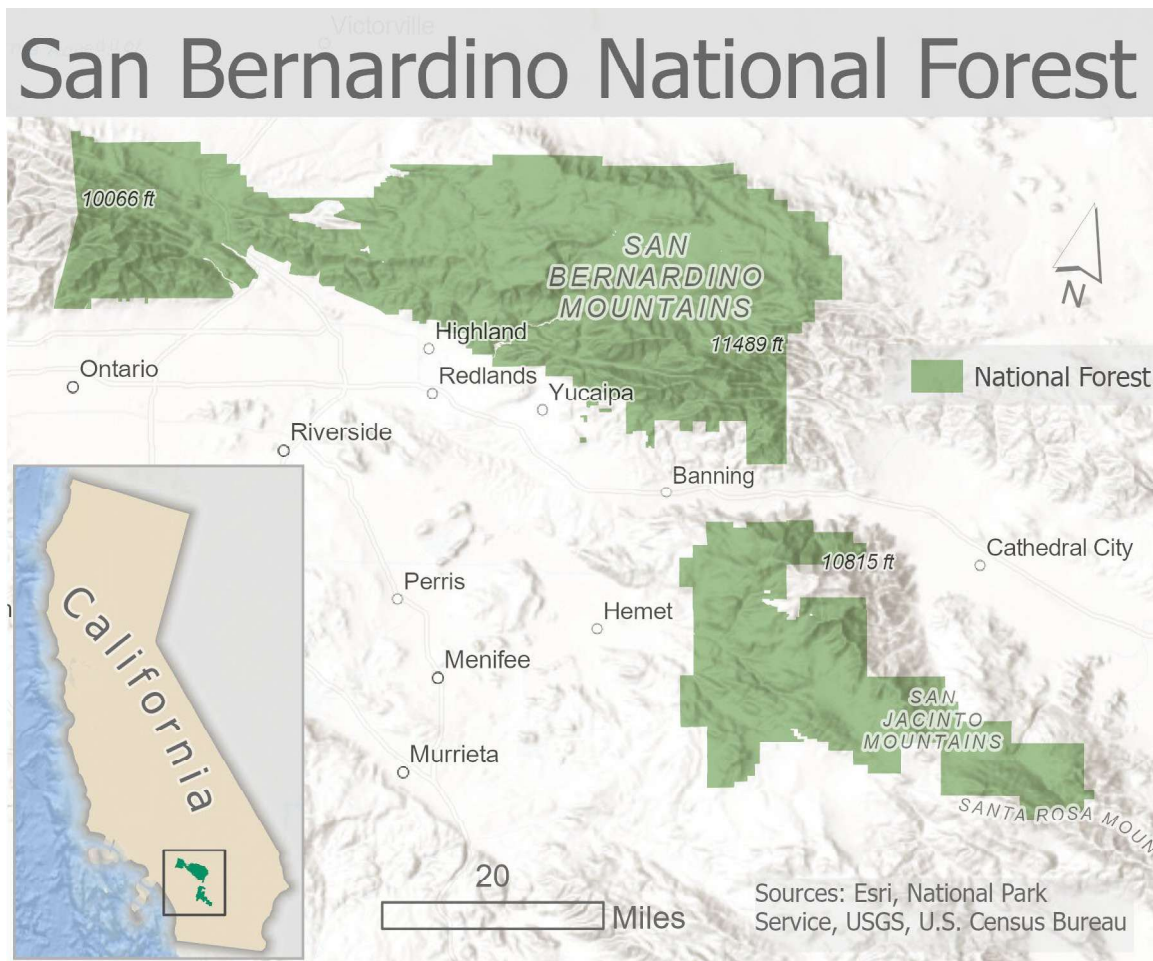


Figure 1-1: Reference map of the San Bernardino National Forest showing its geographic location within Southern California. The map provides spatial context for the study area, including major nearby cities, transportation corridors, and the extent of the forest boundary.

The Line Fire

On the evening of September 5th, 2024, a fire began at Baseline Road and Alpine Street in Highland, California. Within hours, it spread into the northern San Bernardino National Forest (CAL FIRE, 2024). Two days later, the Governor of California declared a state of emergency due to the rapid spread of what became known as the Line Fire.

The fire ultimately burned nearly 44,000 acres and was fully contained on December 23rd, 2024. The Line Fire is used as a case study for this project.

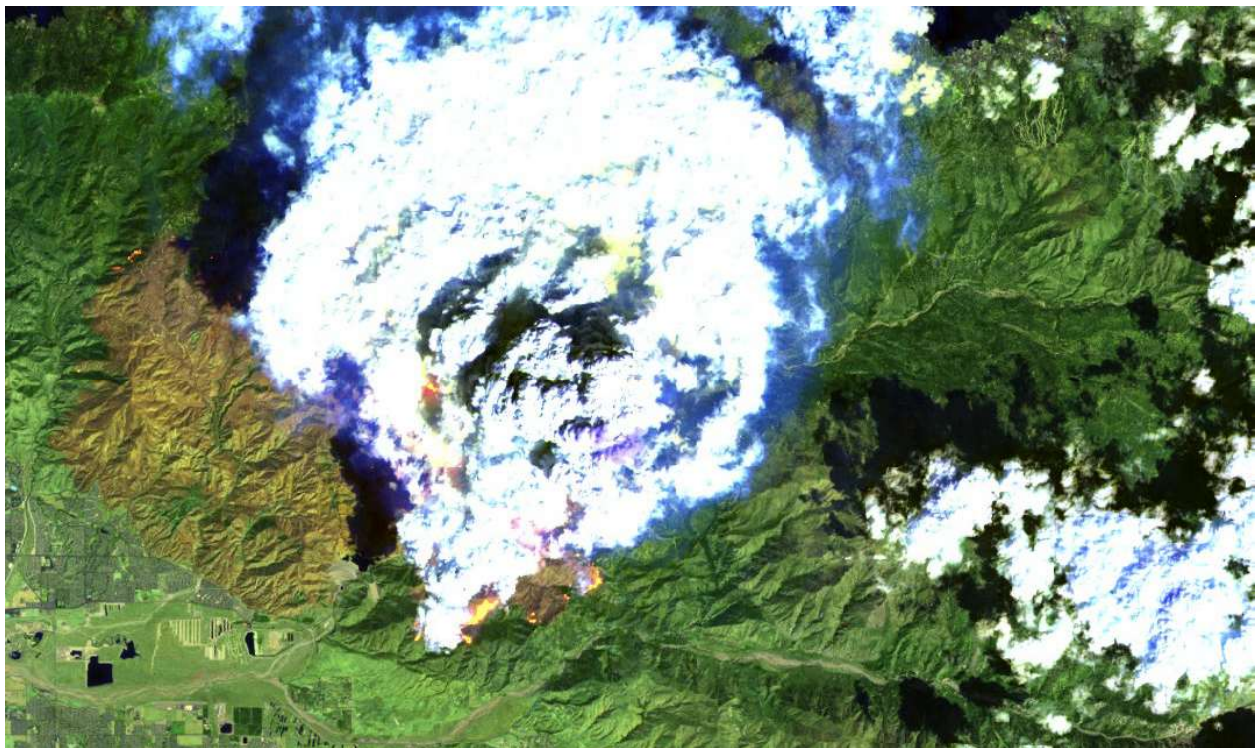


Figure 1-2: False-color composite satellite image of the Line Fire captured on September 9, 2024, using Landsat 8 imagery processed in Google Earth Engine. The composite uses shortwave infrared and red bands (7-6-4), where active fire appears in bright orange, smoke plumes in white to blue, and burned vegetation in dark brown.

Variable Selection

Input variables for wildfire hazard calculation were based on organizational models. These include the Wildland Fire Potential Index (USGS), the European Wildfire

Risk Assessment model (Joint Research Center of the EU), and the CLIMAAX handbook. All these frameworks use similar approaches. They estimate wildfire hazard based on a combination of meteorological, terrain, and vegetation factors that affect fire behavior.

For this project, variables were selected for their relevance to dry and dead fuel, fire spread, and atmospheric drought. The final list includes precipitation, wind speed, NDVI, slope, and vapor pressure deficit (VPD). VPD measures atmospheric *moisture demand*; higher values indicate drier conditions. The link between greater wildfire burn severity and higher VPD is well established (J. T. Abatzoglou & Williams, 2016). Vegetation condition is measured using median composite values of the *lowest NDVI* to reduce seasonal effects, reflecting peak vegetation stress before the fire-prone period.

The Temporal Element of Fire Modeling

Research on the temporal element in wildfire risk and hazard studies varies. Some analyze historical meteorological data and long-term projections. Others do not mention a temporal dimension. The CLIMAAX climate risk assessment framework defines the Fire Weather Index (FWI) as an average fire danger value based on previous seasonal meteorological conditions (Climaax, 2026).

For this case study, the analysis period is the pre-fire season before the Line Fire, from October 2023 to July 2024. Meteorological variables are combined into a composite image and normalized to a common scale. This produces a hazard layer that reflects conditions in the 2024 pre-fire season for the study area.

Google Earth Engine

A comprehensive script was developed in Google Earth Engine to automate the collection and preprocessing of geospatial datasets. The minimal requirements for the

end user are to provide a study area of interest (AOI) polygon and the start and end dates for the desired analysis period. Coefficient weights can be adjusted near the beginning of the script if desired. Optionally, the user may include the name of a Google Drive folder for outputs. Several helper functions are included to assist with common preprocessing steps, such as cloud masking, variable calculations, and value normalization.

Data

For this project, a polygon defines the AOI to include the Line Fire perimeter and nearby areas. Unburned areas are included for comparison during model validation procedures. All data are clipped to the AOI to optimize processing time and memory.

World land cover data from the European Space Agency were used to create a burn mask for forest, shrublands, grasslands, and croplands. Applying the burn mask before normalization reduces potential bias from climate and vegetation variables in non-burnable areas.

Table 1 summarizes the variables derived from various open-source remote sensing products.

Dataset	Source	Resolution	Variable	Processing
WorldCover	European Space Agency	10 m	Burn mask	Binary reclassification
Landsat 8 and 9 Collection 2 Tier 1	United States Geological Survey	30 m	NDVI	Cloud masking; median NDVI composite
Shuttle Radar Topography Mission	NASA Jet Propulsion Laboratory	~ 30 m	Slope	DEM reprojected to EPSG:32611 and slope calculated
gridMET	University of California, Merced & University of Idaho	~ 4 km	Wind speed	Mean composite from daily images
Daymet Version 4	Oak Ridge National Laboratory	1 km	Precipitation, Vapor Pressure Deficit	Cumulative precipitation, VPD calculated from temperature and vapor pressure

Table 1: Summary of remote sensing datasets acquired through Google Earth Engine for wildfire hazard modeling, including data sources, spatial resolution, and their respective roles in deriving hazard variables.

Landsat 8 and 9 Collection 2, Tier 1 satellite images are atmospherically corrected. A cloud mask is applied before calculating NDVI, and a minimum NDVI composite is generated.

Composite images of wind speed, precipitation, and VPD were generated using the gridMET and Daymet (Version 4) datasets. VPD was calculated from minimum and

maximum temperatures and atmospheric vapor pressure, and daily values were averaged over the period.

The script's most complex function, `getTransformList`, extracts projection transformation information from a referenced Landsat image within the AOI. Because USGS uploads Landsat imagery in different UTM projections, the script retrieves the transformation from one image and applies it to the *slope* variable.

All outputs are reprojected to UTM Zone 11N (EPSG: 32611) during calculation and export. This matches the projection of the first Landsat image collected by the script.

Wildfire Hazard Index Calculation

After processing, all five variables are combined into a *hazard composite image*. Each variable is normalized to 0–1, where higher values indicate greater fire hazard. Precipitation and NDVI, which represent dry, dead fuels when low, use an inverse function, so that low or negative values reflect high hazard. All variables are combined with a weighted linear equation to generate the Wildfire Hazard Index (WHI):

$$WHI = (0.30 \times P^{-1}) + (0.10 \times W) + (0.15 \times VPD) + (0.10 \times NDVI^{-1}) + (0.35 \times S)$$

Where:

Variable	Description
P^{-1}	Inverse of normalized total precipitation
W	Normalized mean wind speed
VPD	Normalized Vapor Pressure Deficit
$NDVI^{-1}$	Inverse of normalized NDVI
S	Normalized Slope

Table 2: Variable abbreviations incorporated into the weighted linear combination equation used to generate the Wildfire Hazard Index (WHI).

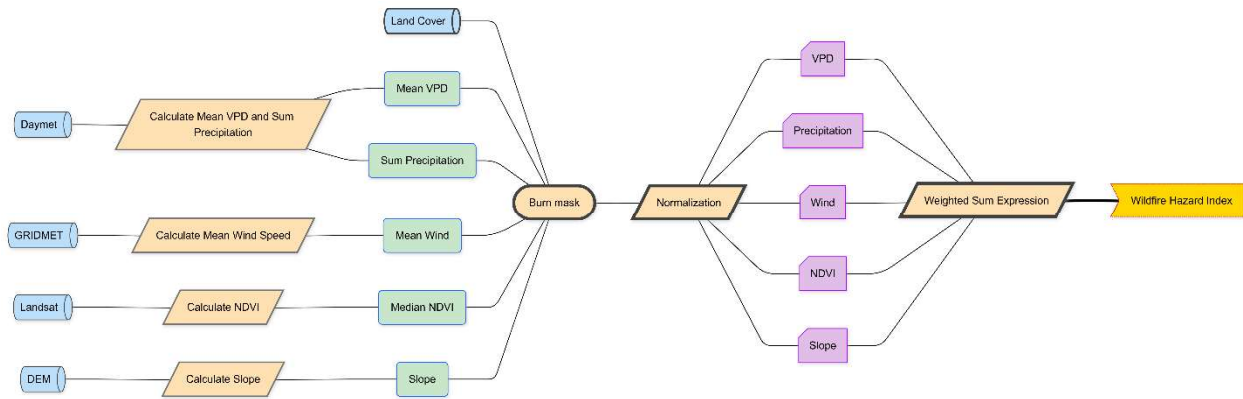


Figure 1-3: Workflow diagram showing the process in Google Earth Engine to generate the Wildfire Hazard Index (WHI). Input variables from remote sensing datasets are normalized and combined using weighted linear methods. The result is a continuous raster representing relative wildfire hazard across the study area.

Greater weight was given to precipitation and slope-hazard variables. These influence the fuel dryness and the rate of spread across landscapes. Weights are based on their importance to the region. In Southern California, drought and steep terrain often lead to rapid fire spread, so these factors are weighted accordingly. Because wildfire dynamics differ by climate, vegetation, and season, this weighting scheme is context-specific.

dNBR

An additional script created a composite Differenced Normalized Burn Ratio (dNBR) raster for the same AOI. Image collections from Landsat 8 and 9 were used to calculate the NBR for both pre- and post-fire conditions. Median NBR values reduce the influence of cloud contamination and seasonal variability. The dNBR raster is created by subtracting post-fire NBR from pre-fire NBR.

The pre-fire composite uses images from July to September 1, 2024. The post-fire composite uses images from September 1, 2024, to January 31, 2025. The resulting burn severity raster is used to evaluate the wildfire hazard model's accuracy.

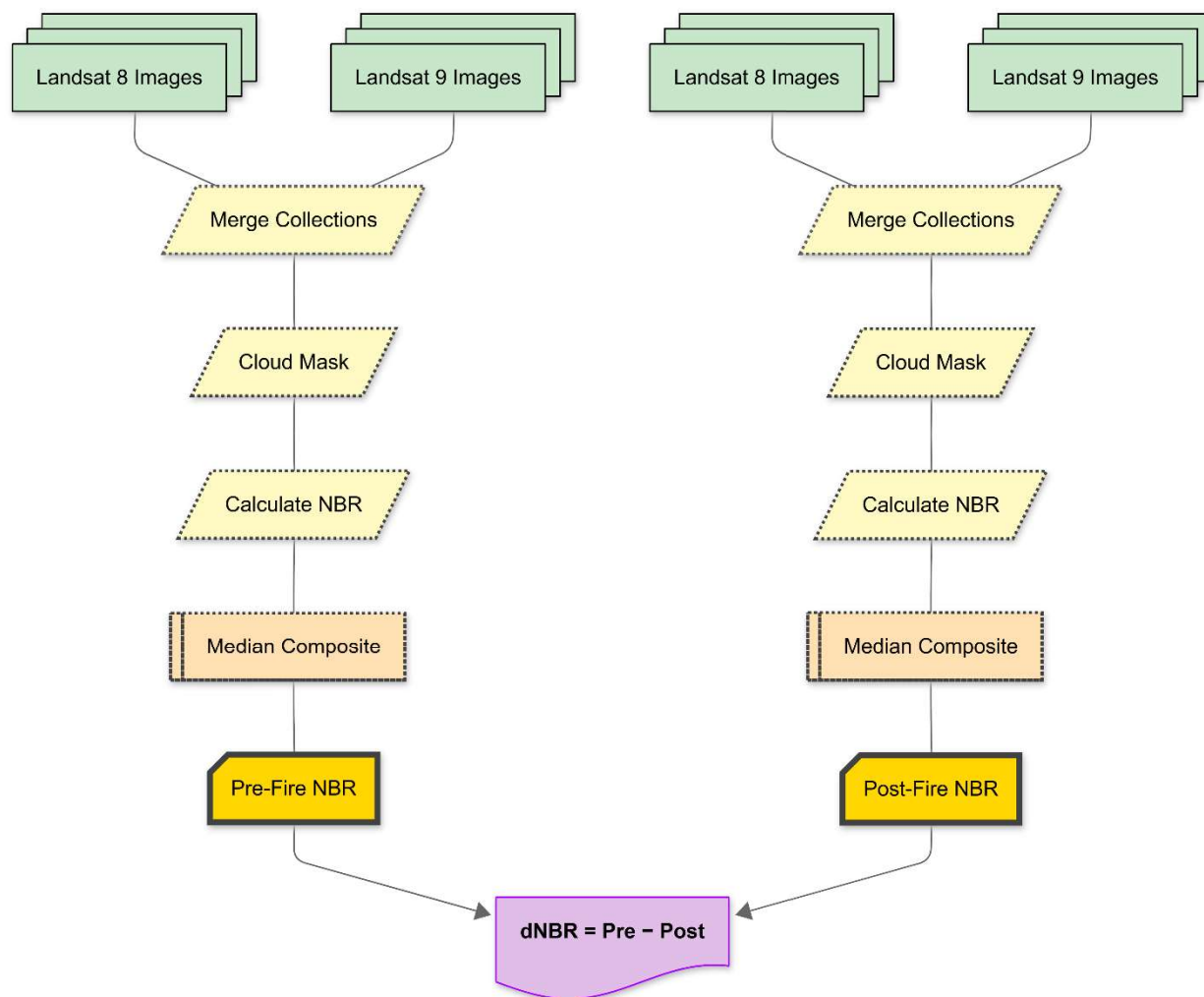


Figure 1-4: Workflow diagram showing the generation of differenced Normalized Burn Ratio (dNBR) using Landsat 8 and Landsat 9 imagery within Google Earth Engine. Pre-fire and post-fire median composites are used to calculate NBR, which are then differenced to produce a burn severity raster for comparison with the modeled hazard.

Accuracy Assessment

Historical wildfire perimeter data were retrieved from the California Department of Forestry and Fire Protection. Using QGIS software, the Line Fire perimeter was saved as a new shapefile and used to create burned and unburned area polygons within the extent of the exported WHI raster. Spatial statistics were reviewed, including a histogram of WHI values, giving insights into the distribution of values across the study area. The model was assessed using two methods:

1. Statistical significance of the difference in WHI values in burned and unburned areas
2. Spatial comparison of WHI and burn severity (dNBR) values inside the fire perimeter

For the first accuracy measure, two sets of 1,000 random sample points were generated within each polygon—one set in burned areas and one in unburned areas. WHI values from each set of sample points were extracted and analyzed using RStudio statistical software.

For the second accuracy measure, an additional 1,000 random sample points were generated within the Line Fire perimeter. Both WHI and dNBR rasters were stacked, and pixel values were extracted from the stack. The rasters were reclassified into four categories: low, moderate, high, and very high, based on their respective ranges. To assess the performance of the estimated hazard relative to burn severity, a confusion matrix is created, and overall accuracy metrics are calculated. This test investigated the overall agreement between the predicted hazard from the WHI and the “ground-truth” observed in the dNBR values. Random sampling for both accuracy measures allows for well-distributed observations.

All GEE and R scripts used for data processing, plotting, and statistical analysis are available in the project’s GitHub repository at <https://github.com/hrobbins-gis/wildfire-hazard-gee>. Metadata for all spatial datasets and Google Earth Engine assets used in this analysis are provided in the Appendix of this report.

RESULTS

The hazard modeling workflow developed in Google Earth Engine (GEE) was initially tested across multiple smaller areas, confirming that consistent and logical output surfaces were generated. Through this process, a computational threshold was identified: polygons exceeding approximately 1,000 km² resulted in timeouts during intermediate computations of variables and hazard raster's.

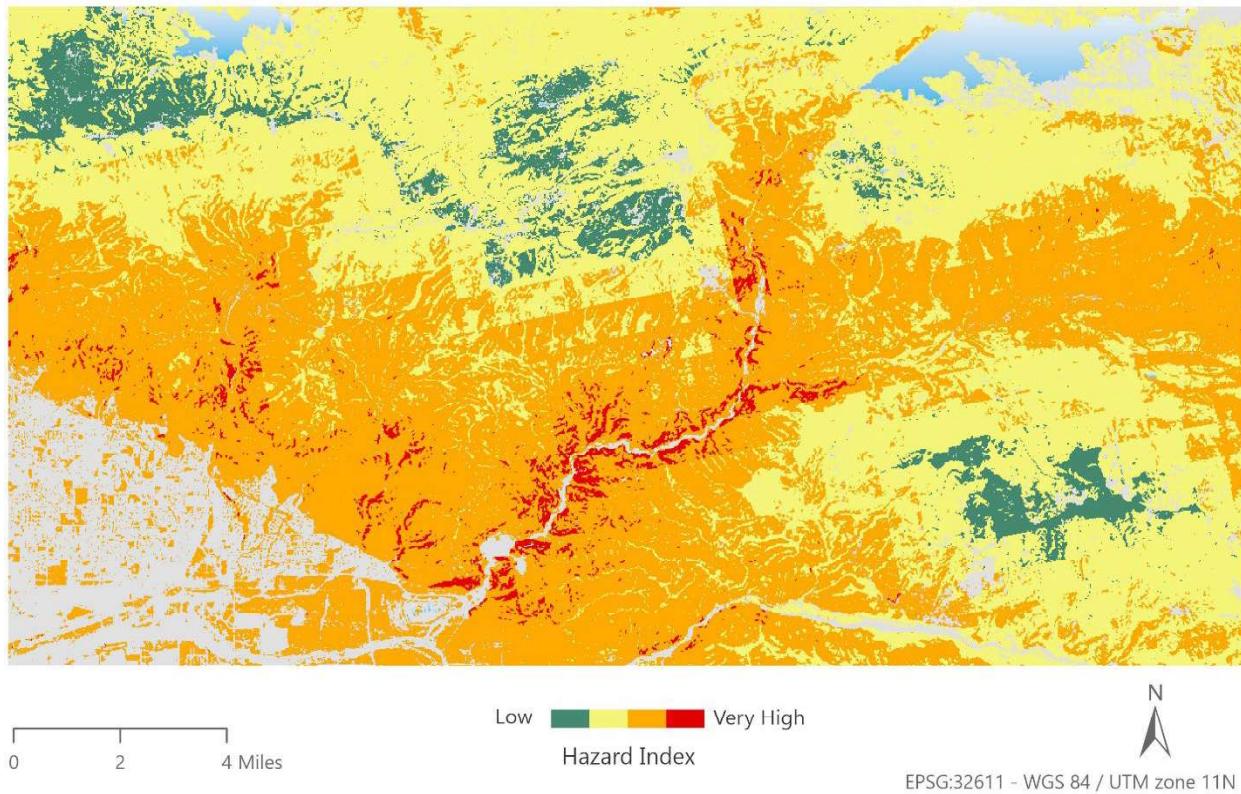


Figure 2-1: Final Wildfire Hazard Index (WHI) raster displaying the spatial distribution of modeled wildfire hazard across the study area. Values range from low (0) to very high hazard (1), with higher values indicating areas of increased susceptibility.

The WHI (Figure 2-1) ranges from 0 to 1. Non-burnable surfaces are masked and shown in gray, while water bodies are shown in light blue. Low (0-0.24) and moderate (0.25–0.49) hazard values are concentrated in the southeastern and much of the northern portions of the study area, visualized in green and yellow, respectively. In

contrast, high hazard values (0.50–0.74), shown in orange, are observed in the southwestern region along the wildland–urban interface near Highland, and extend northward into forested areas and eastward across the study area. “Very High” hazard values (>0.74) appear red on the map and are specifically concentrated along the Santa Ana River corridor. This spatial pattern illustrates the variation in modeled hazard throughout the region.

These high-hazard areas correspond to zones characterized by relatively low precipitation, elevated moisture deficit, and steeper terrain within the study area.

Summary statistics of the WHI distribution show a median of 0.49, a standard deviation of 0.14, and a kurtosis of 2.41. The histogram in **Figure 2-2** illustrates that WHI values are not uniformly distributed. Slight negative skewness indicates a greater proportion of high-hazard values. These results show the non-uniform distribution of wildfire hazard across the study area.

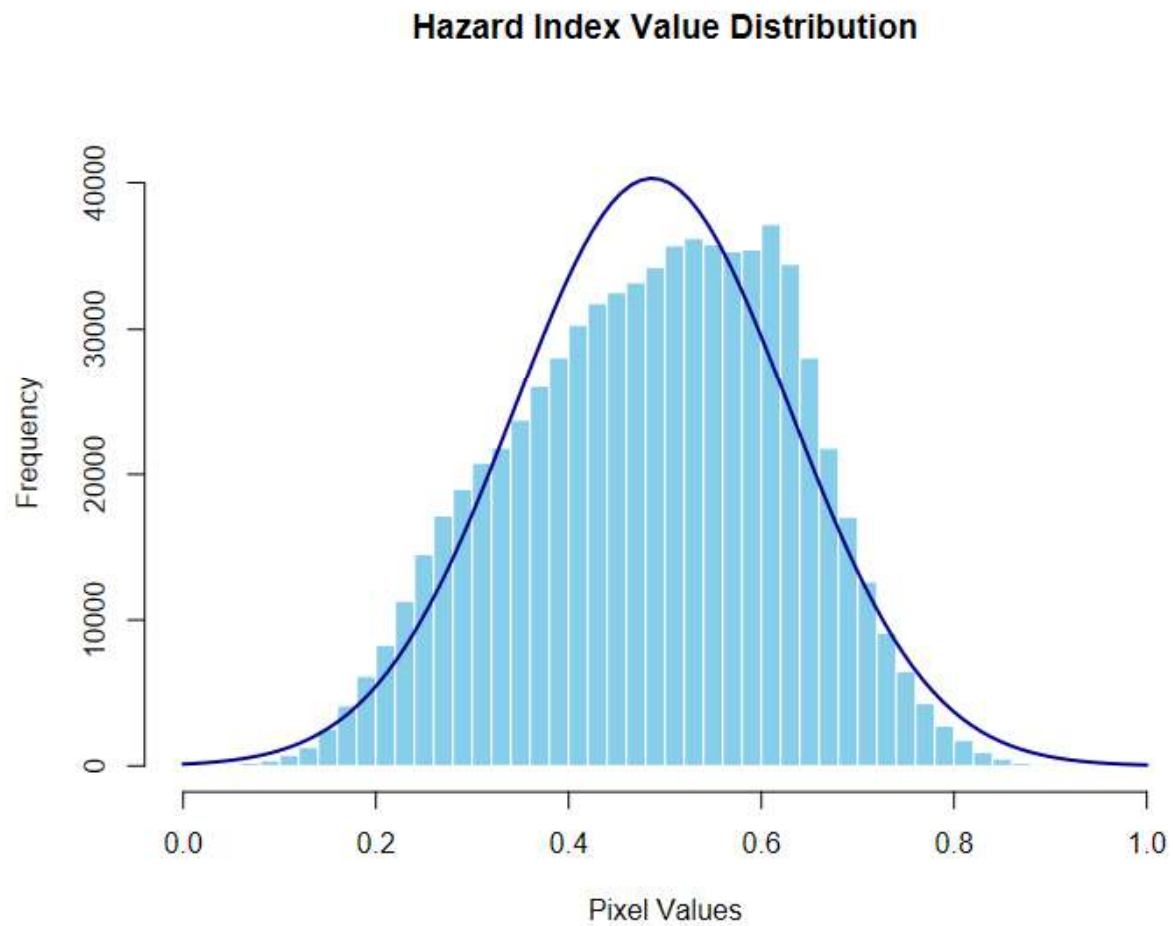


Figure 2-2: Distribution of WHI values across the study area shown as a histogram, with an overlaid normal distribution curve for comparison. This figure illustrates the overall spread, central tendency, and skewness of hazard values.

Accuracy Assessment 1

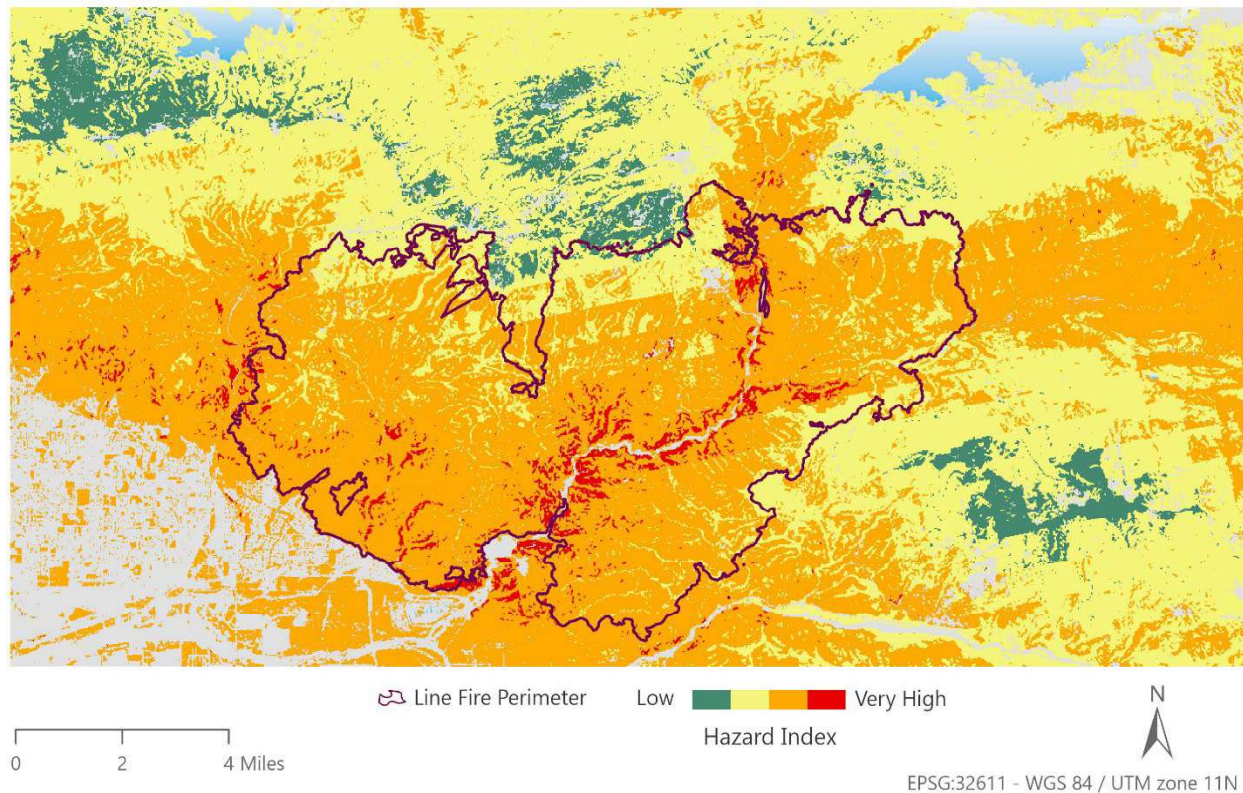


Figure 2-3: Wildfire Hazard Index (WHI) raster overlaid with the perimeter of the Line Fire. Areas of highest hazard (0.75–1.0) are concentrated within the central portion of the fire perimeter, indicating spatial agreement between modeled hazard and observed fire extent.

To evaluate the hazard model's performance, statistical analyses were completed to compare WHI values inside the observed fire perimeter with those in the surrounding unburned landscape. This strategy provides a direct assessment of model predictive ability.

WHI values were not normally distributed, so a non-parametric test was used. The Wilcoxon rank-sum test (also called the Mann–Whitney U test) compares two independent samples by ranking all observations. It tests if one group's values are

higher than the other's. This method is suitable for comparing burned and unburned sample points, as it requires no assumptions of data distribution.

The null hypothesis (H_0) states that WHI values within the fire perimeter are not higher than those outside the perimeter. The alternative hypothesis (H_1) states that the values inside are higher. The test result ($W = 775,882$; $p < 0.001$) shows that WHI values within the perimeter are significantly higher. This suggests the model clearly identifies areas of high wildfire hazard.

A boxplot created in R provides a visual summary of differences in sample distributions. As highlighted in Figure 2-4, the median WHI value within the burned area is 0.59, compared to 0.45 in unburned areas.

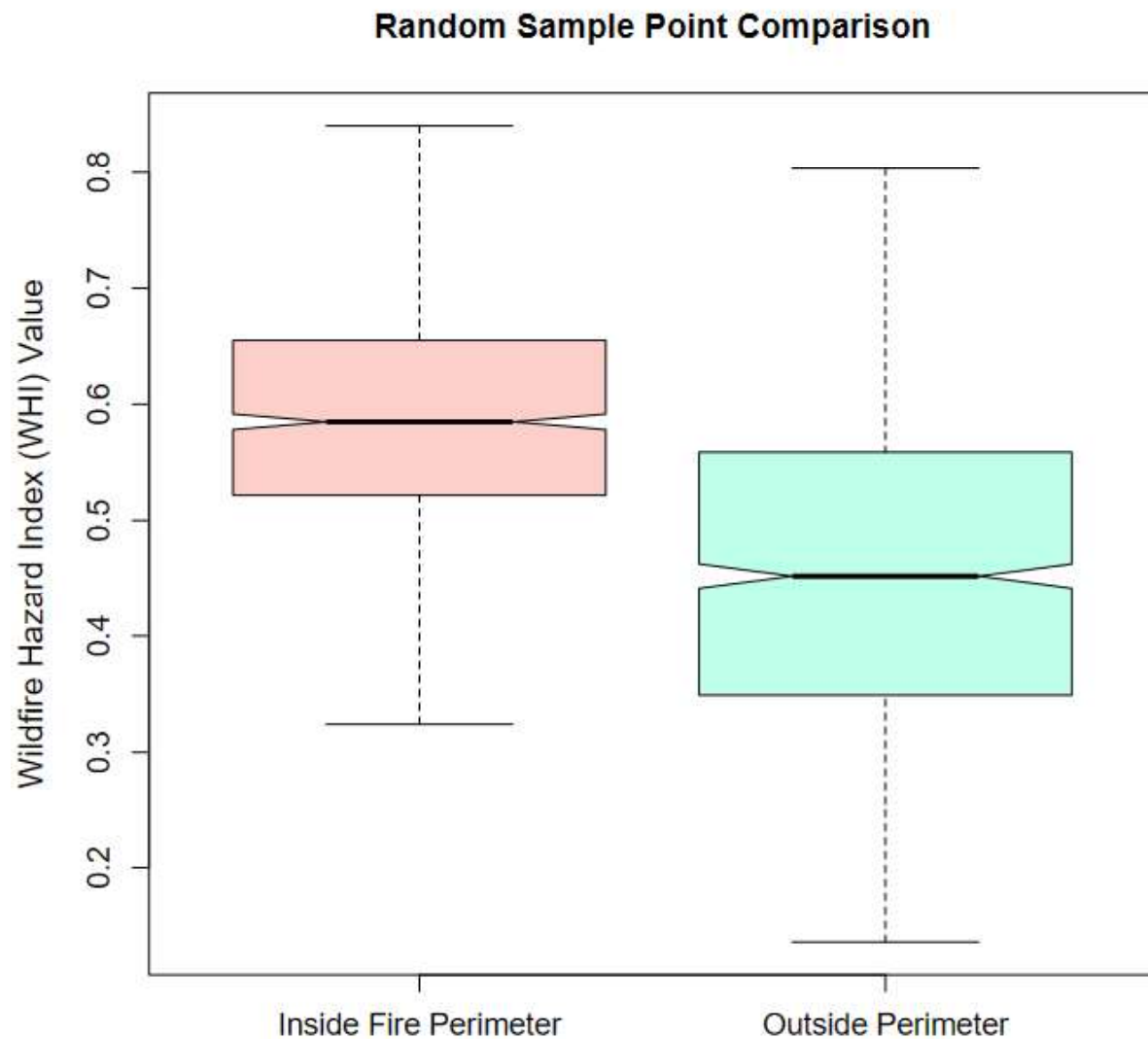


Figure 2-4: Boxplot comparing the distribution of WHI sample values inside and outside the Line Fire perimeter. The figure illustrates differences in median, spread, and variability between burned and unburned areas.

Additionally, the distribution of values within the fire perimeter shows a larger overall range of values and higher maximum values than in the unburned samples. These patterns are visually supported by the figure, which highlights that high WHI values are more frequently associated with areas that burned during the Line Fire.

Large sample sizes can yield statistically significant p-values even if differences are small. Because the W statistic also depends on sample size, its magnitude should be interpreted carefully (Bobbitt, 2021). For this reason, an effect size was calculated to measure the practical significance of the observed difference and provide additional insight alongside statistical significance.

Effect size describes the importance of a relationship or difference, independent of sample size. In this study, the rank-biserial correlation—which measures the strength of association between two groups—was used as a complementary measure to the Wilcoxon rank-sum test. Rank-biserial values range from 0 (no difference) to 1 (complete separation between groups).

The effect size ($r = 0.55$) shows a moderate effect. This means a clear, meaningful distinction exists between WHI values inside and outside the fire perimeter. The model effectively estimates high wildfire hazards within known observed events, beyond statistical significance alone.

To further explore the model's effectiveness, a confusion matrix was created to review the relationship between the modeled hazard and burn severity.

Accuracy Assessment 2

The differenced Normalized Burn Ratio (dNBR) raster (Figure 2-5) shows higher burn severity along the northern and eastern sections of the Line Fire perimeter. The dNBR values clearly delineate and support the shape of the mapped fire perimeter. More severe burns are concentrated in the northeast. Lower severities are found in the southwest near Highland.

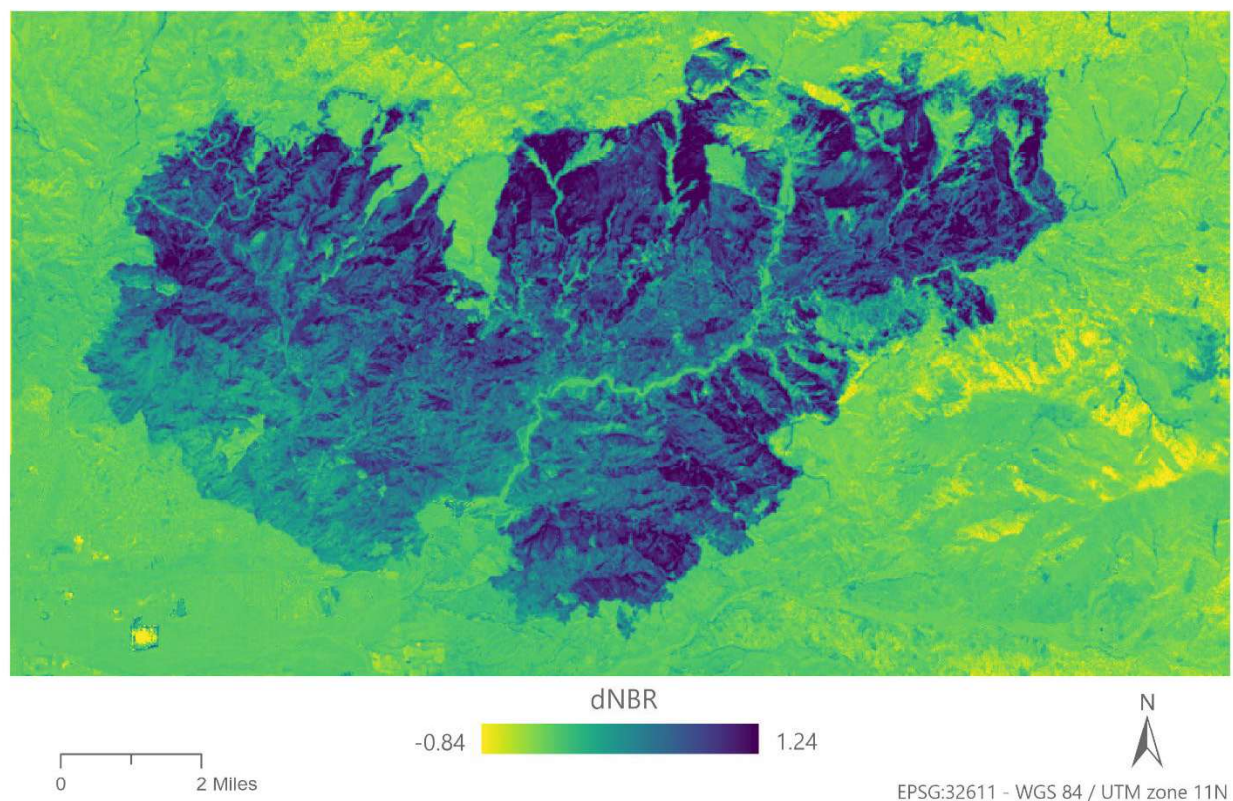


Figure 2-5: Differenced Normalized Burn Ratio (dNBR) raster generated in Google Earth Engine, representing burn severity across the study area. Higher values correspond to increasingly severe burn impacts, while lower or negative values indicate unburned or minimally affected areas.

The WHI and dNBR rasters were clipped to the Line Fire perimeter for comparative evaluation. An additional 1,000 random sample points were generated to extract *paired* WHI and dNBR values. Through this direct comparison, dNBR is used as a proxy for *observed* burn severity, enabling subsequent categorical analysis with the modeled hazard.

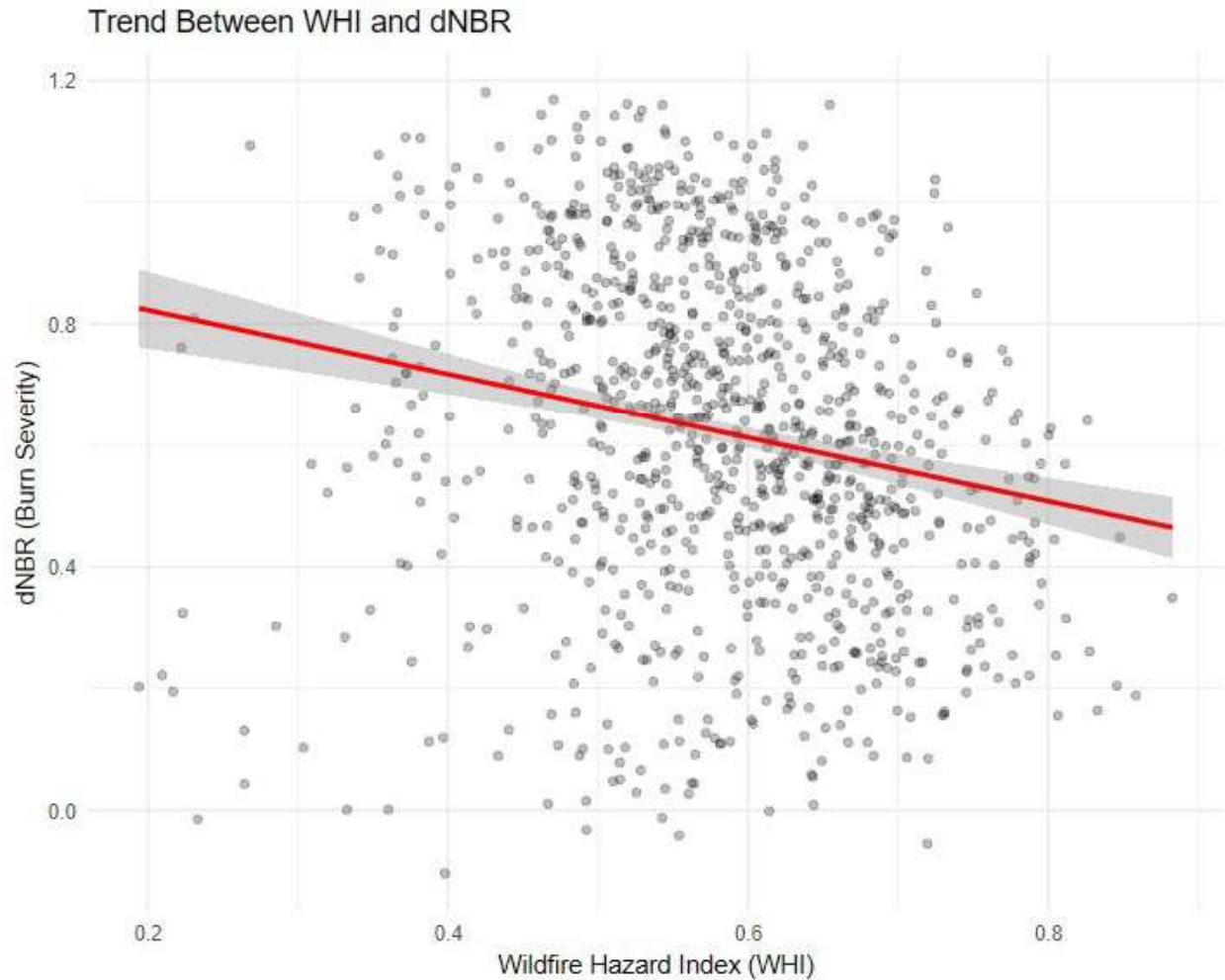


Figure 2-6: Scatterplot showing the relationship between dNBR and Wildfire Hazard Index (WHI) sample values, with an overlaid trend line indicating the direction and strength of correlation between modeled hazard and observed burn severity.

Figure 2-6 displays a scatter plot of WHI and dNBR values with a fitted regression line which shows a weak negative trend and considerable variation in the midrange. This suggests that higher modeled hazard areas do not always match higher observed burn severity.

To further evaluate this relationship, both WHI and dNBR values were reclassified into four equal-interval categories: Low, Moderate, High, and Very High. This reclassification allows for simpler categorical comparisons and allows for visual

and statistical analysis. Thresholds and original value ranges are in Appendix Table AP-1.



Figure 2-7: Boxplots illustrating the distribution of dNBR values across equal-interval WHI categories (Low, Moderate, High, Very High). This figure evaluates how burn severity varies across increasing hazard classes.

A downward trend in median dNBR values is evident across the “Moderate” to “Very High” WHI categories. This supports the inverse relationship observed in the scatter plot. The “Low” class is more variable due to fewer samples. The “Moderate” and “High” classes include most sampled points, showing the main WHI distribution.

The “High” WHI class has many observations and a wide range of values, showing that burn severity varies greatly in this group. Still, the median and upper quartile values are lower than those in the “Moderate” class, supporting the overall

inverse trend. The “Very High” class continues this trend, with even lower median dNBR values.

Overlap exists across all classes, a pattern expected in complex environmental systems. Despite this overlap, the progressive shift in distributions provides evidence that WHI captures landscape conditions that differ from—but are related to—observed burn severity.

To further explore the relationship, a confusion matrix was created to visualize and calculate the agreement between predicted hazard and observed burn severity (Figure 2-8).

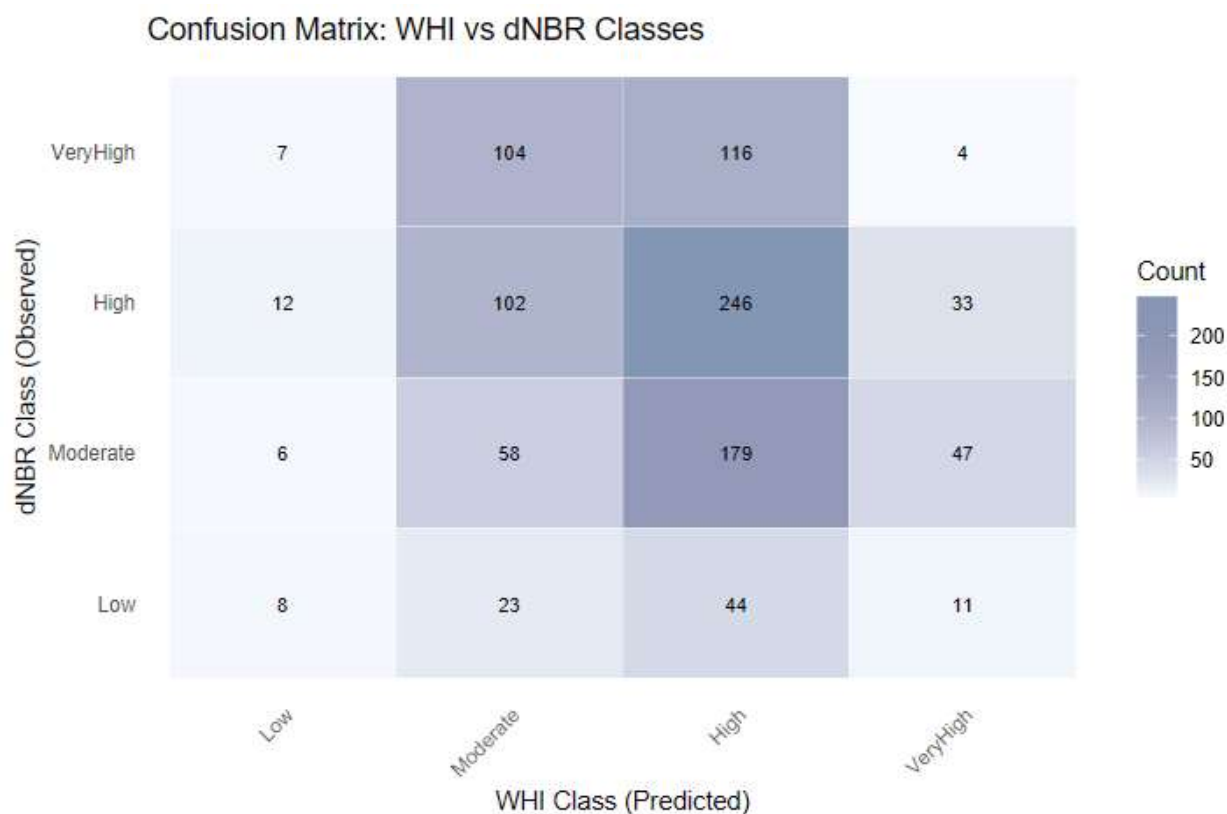


Figure 2-8: Confusion matrix comparing categorized Wildfire Hazard Index (WHI) classes with corresponding burn severity classes derived from dNBR. The matrix quantifies agreement and misclassification between predicted hazard and observed fire effects.

Class	Producer's Accuracy	User Accuracy
Low	9%	24%
Moderate	20%	20%
High	63%	42%
Very High	2%	4%
Overall Accuracy		32%

Table 3: Producer's accuracy, user's accuracy, and overall accuracy derived from the confusion matrix comparing Wildfire Hazard Index (WHI) classifications and burn severity classes.

Table 3 presents the producer's, user's, and overall accuracies from the confusion matrix comparing hazard and burn severity classes. The producer's accuracy is the chance of correctly mapping reference pixels in each class. The user's accuracy shows how reliable the mapped classes are. Overall accuracy is the proportion of correctly classified samples across all categories and measures the model's general performance in predicting burn severity.

The most significant agreement occurs in the "High" classification, with producer and user accuracies of 63% and 42%, respectively. The difference between the two accuracies indicates that many true high-burn-severity pixels were correctly identified. However, the model also assigns non-high-severity areas to this class. All other classes display lower levels of agreement. Overall classification accuracy spanning all categories is 32%. This implies that, although the model captures aspects of wildfire hazard, it does not strongly predict spatial patterns of burn severity.

CONCLUSION

Model Effectiveness/Interpretation

Wildfires are unpredictable in terms of when and where they start. However, factors like fuel, moisture, and topography are more predictable and can be modeled to estimate fire behavior. Mitigation and suppression strategies alter these relationships as they change fire dynamics.

Google Earth Engine streamlines data processing, improving reproducibility and accessibility. In the Line Fire case study, the workflow generated a hazard surface from selected variables and a time frame. Statistical analysis showed higher WHI values inside the Line Fire perimeter than outside, indicating the model reliably captures pre-fire hazard.

The workflow estimates wildfire hazard distribution well but does not accurately predict burn severity. Estimated hazard correlated weakly with actual burn severity. This shows complex local factors like weather, fuel availability, and suppression efforts greatly affect fire impact. For example, prescribed burns in the Angelus Oaks Understory Burn Project reduced fire spread in the Line Fire (CAL FIRE, 2024), adding unpredictability not captured by the model.

The project's WHI result is most effective for identifying at-risk areas and supporting preparedness and risk mitigation, rather than for predicting specific fire impacts. Results show that the proposed workflow effectively estimates relative wildfire hazard but is not suitable for predicting burn severity. The model is best applied to identify areas of high susceptibility based on environmental conditions.

Limitations

Google Earth Engine is powerful for large-scale analysis, but it has limits. Processing times increase with study area size, and large polygons (over 1,000 km²) may cause computation or export timeouts. Some global datasets have limited spatial or temporal resolution, reducing model capability.

On-the-fly reprojection in GEE can cause subtle visual artifacts, like offset or diagonal pixels, but pixel geolocation remains accurate (Gorelick et al., 2017). This differs from desktop GIS workflows and may require extra user validation.

Input data are constrained by a tradeoff between spatial and temporal resolution. Climate datasets offer high temporal but coarse spatial resolution (approximately 1-4 km), whereas Landsat provides finer spatial detail (30 m) but lower temporal frequency. This mismatch may reduce output effectiveness, particularly in local assessments where higher spatial resolution is preferred.

Variable selection posed extra limitations. Wind, which drives fire spread, was not included in a short-term, event-specific way. Future work should use short-term wind conditions before fire events instead of relying on seasonal averages. Wind data are also unavailable at finer spatial resolutions; better climate products would improve fire hazard modeling.

Inaccuracies in the burnable area mask may also have introduced minor errors into hazard calculations. For example, some non-burnable areas (e.g., water edges) may have been inadvertently included during normalization and as model inputs, affecting the hazard surface. This emphasizes the need for meticulous data preparation and the learning curve involved in transitioning to code-based environments like GEE.

Strengths and Future Work

This project proves automated susceptibility modeling works with cloud-based geospatial analysis. Google Earth Engine allows fast data processing that would be too time-consuming with desktop GIS.

The provided workflow demonstrates the potential for repeatable susceptibility modeling. As open-source remote sensing datasets continue to expand, similar methodologies can be adapted to a broad range of environmental applications beyond wildfire science. Cloud-based geospatial computing is expected to play an increasingly prominent role in the future of GIS.

A practical direction for future research is to develop a web application that allows users to define study areas and generate wildfire hazard outputs without coding. This development could improve both accessibility and usefulness for a broad range of users. Further research should also test and optimize input variable selection and weighting across different geographic settings to improve model performance. Comparative studies assessing the effectiveness of the hazard index across different ecoregions would help clarify how region-specific variables affect model performance and encourage tailored hazard assessments.

Combining GEE with desktop GIS is promising. GEE efficiently gathers and processes data, while desktop GIS provides greater control over reprojection, data cleaning, and map products. Using GEE for initial processing and desktop GIS for analysis and visualization can improve both accuracy and usability.

Overall, this project provides a potential steppingstone for future wildfire hazard studies. Further development can produce more practical and efficient tools and expand environmental modeling applications.

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APPENDIX

METADATA TABLES

Table AP-1

Dataset:	Fire Perimeters File Geodatabase
Last Updated:	April 2025
Owner:	The Fire and Resource Assessment Program (FRAP)
Description:	Historical fire perimeters across California within public and private lands.
Coordinate System:	North American Datum 1983
Projection:	California Albers EPSG:3310
Type of Geometry:	Vector

Table AP-2

Dataset:	ESA World Cover
Last Updated:	October 2021
Owner:	European Space Agency
Google Earth Asset ID:	"ESA/WorldCover/v100"
Description:	Land cover classification based on Sentinel 1 and Sentinel 2 satellites at 10 m resolution.
Coordinate System:	World Geodetic System 1984 (WGS 1984)
Projection:	Geographic Coordinate System EPSG: 4326
Type of Geometry:	Raster

Table AP-3

Dataset:	USGS SRTM Digital Elevation Model
Last Updated:	February 2020 (Collection updates; original mission 2000)
Owner:	United States Geological Survey
Google Earth Asset ID:	"USGS/SRTMGL1_003"
Description:	Global digital elevation model from the Shuttle Radar Topography Mission (SRTM), providing near-global elevation coverage at ~30 m spatial resolution.
Coordinate System:	World Geodetic System 1984 (WGS 1984)
Projection:	Geographic Coordinate System EPSG: 4326
Type of Geometry:	Raster

Table AP-4

Dataset:	Landsat 8 (Collection 2, Tier 1)
Last Updated:	Ongoing (Collection 2 released 2021)
Temporal Resolution:	16 days
Owner:	United States Geological Survey / NASA
Google Earth Asset ID:	"LANDSAT/LC08/C02/T1_L2"
Description:	Surface reflectance imagery from Landsat 8 Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS), providing multispectral data at 30 m resolution for land monitoring and analysis.
Coordinate System:	World Geodetic System 1984 (WGS 1984)
Projection:	Universal Transverse Mercator (UTM, zone dependent)
Type of Geometry:	Raster

Table AP-5

Dataset:	Landsat 9 Surface Reflectance (Collection 2, Tier 1)
Last Updated:	Ongoing (Collection 2 released 2021; Landsat 9 launched 2021)
Temporal Resolution:	16 days
Owner:	United States Geological Survey / NASA
Google Earth Asset ID:	"LANDSAT/LC09/C02/T1_L2"
Description:	Surface reflectance imagery from Landsat 8 Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS), providing multispectral data at 30 m resolution for land monitoring and analysis.
Coordinate System:	World Geodetic System 1984 (WGS 1984)
Projection:	Universal Transverse Mercator (UTM, zone dependent)
Type of Geometry:	Raster

Table AP-6

Dataset:	Daymet Version 4 Daily Surface Weather Data
Last Updated:	Annual updates (Version 4 released ~2020, ongoing updates)
Temporal Resolution:	Daily
Owner:	Oak Ridge National Laboratory
Google Earth Asset ID:	"NASA/ORNLDAYMET_V4"
Description:	Daily gridded meteorological dataset providing surface weather parameters including temperature, precipitation, and vapor pressure at ~1 km spatial resolution across North America.
Coordinate System:	Lambert Conformal Conic (native projection)
Projection:	Custom Lambert Conformal Conic projection (Daymet native)
Type of Geometry:	Raster

Table AP-7

Dataset:	gridMET (Gridded Surface Meteorological Dataset)
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Last Updated:	Ongoing (daily updates)
Temporal Resolution:	Daily
Owner:	University of Idaho
Google Earth Asset ID:	"IDAHO_EPSCOR/GRIDMET"
Description:	High temporal resolution meteorological dataset providing daily surface variables such as precipitation, temperature, wind speed, and evapotranspiration at ~4 km spatial resolution across the United States.
Coordinate System:	World Geodetic System 1984 (WGS 1984)
Projection:	Geographic Coordinate System EPSG: 4326
Type of Geometry:	Raster

RECLASSIFICATION TABLES

Table AP-8: Equal interval classification thresholds applied to randomly sampled values of the Wildfire Hazard Index (WHI) and differenced Normalized Burn Ratio (dNBR).

Class	dNBR Range	WHI Range
Low	-0.10 – 0.21	0.19 – 0.35
Moderate	0.22 – 0.53	0.36 – 0.53
High	0.54 – 0.85	0.54 – 0.70
Very High	0.86 – 1.18	0.71 – 0.88